Procedure

Testing of a Semiconductor diode/ zener diode using Analog multimeter:

1. Select X1 Ohm's range and set 0 Ω adjustment.
2. Connect Anode with Black test probe and Cathode with Red test probe. If the meter reading is approximately 7 Ω, the diode is said to be in **good condition.** This is known as ***forward bias testing***.

   If the meter reading shows 0 Ω, diode is said to be ***short.***

   If the meter reading shows ∞ Ω , i.e., the needle does not move, diode is said to be ***Open.***

   If the meter reading shows some reading above 7 Ω , i.e., 100 Ω or 5k Ω, it means the diode has high ***'forward bias resistance'.***

> **Note**
>
> Diodes with the following conditions ***cannot be used*** in circuits.
>
> 1. Open (or)
> 2. Short circuit (or)
> 3. Forward bias resistance becomes high.

3. Connect Anode with Red test probe and Cathode with black test probe. If the meter reading is ∞ Ω, the diode is said to be in ***good*** condition. This is known as ***reverse bias testing***.

   If the meter reading shows 0 Ω, diode is said to be ***short.***

   If meter reading shows some reading that is 300 Ω or 6 k Ω, diode is said to be ***Leaky.***

> **Note**
>
> Leaky Diodes ***cannot be used*** in circuits.

Tabular column: (Draw separate tabular column and do test for each diode)

S.No.	Testing	Meter selector switch position	Anode	Cathode	Meter Reading	Result
1	Forward bias	X 1	- ve probe	+ ve probe		
2	Reverse bias	From X 1 to X 1 K	+ ve probe	- ve probe		
Final Result :						

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